

## Lab 2: Agile Backlog Creation and Sprint Simulation in Jira

### Problem Statement #62 — Food Bank Surplus Redistribution Platform

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### 1. Project Setup

A Scrum project was created in Jira using the Software Development template and the company-managed project type. The project is named Food Bank Surplus Redistribution with key FBSR. The functional requirements identified in Lab 1 (FR-001 to FR-005 and NFR-002) were used as the basis for the Epics and User Stories below.

### 2. Epics

| Key    | Epic                          | Description                                                                                                                                                 |
|--------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FBSR-1 | Surplus Listing Management    | Enable donor partners to post surplus food batches with expiry countdowns and dietary tags, and ensure expired batches leave the claim board automatically. |
| FBSR-2 | Batch Discovery & Reservation | Enable shelter coordinators to find suitable batches and reserve a pickup window with a verifiable claim code.                                              |
| FBSR-3 | Handover & Traceability       | Confirm physical handover of food and maintain a complete audit trail of every batch event.                                                                 |
| FBSR-4 | Trust & Notifications         | Verify shelter legitimacy and alert nearby shelters when perishable food is listed.                                                                         |

### 3. User Stories

Thirteen user stories were written across the four Epics using the As a / I want / So that format. Priorities were assigned by user impact and story points using the Fibonacci scale.

| Key     | Story                              | Epic   | User story                                                                                                                                                    | Priority | Points |
|---------|------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FBSR-5  | 1.1 Post surplus batch             | FBSR-1 | As a donor partner, I want to post a surplus food batch with quantity and description, so that nearby shelters can see what is available.                     | High     | 5      |
| FBSR-6  | 1.2 Expiry countdown timer         | FBSR-1 | As a donor partner, I want each batch to show a live expiry countdown, so that shelters know how long they have to collect it.                                | High     | 3      |
| FBSR-7  | 1.3 Dietary tagging                | FBSR-1 | As a donor partner, I want to tag batches as vegetarian, halal or contains-nuts, so that shelters can match food to their residents' needs.                   | Medium   | 2      |
| FBSR-8  | 1.4 Auto-expire listing            | FBSR-1 | As a shelter coordinator, I want expired batches removed from the claim board automatically, so that I never travel for food that is no longer safe to serve. | High     | 5      |
| FBSR-9  | 2.1 Search by distance             | FBSR-2 | As a shelter coordinator, I want to filter the claim board by distance radius, so that I only see food I can collect before it expires.                       | High     | 3      |
| FBSR-10 | 2.2 Filter by dietary tag and time | FBSR-2 | As a shelter coordinator, I want to filter by dietary tag and time left, so that I can find suitable food quickly.                                            | Medium   | 2      |

| Key     | Story                           | Epic   | User story                                                                                                                                               | Priority | Points |
|---------|---------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| FBSR-11 | 2.3 Reserve pickup window       | FBSR-2 | As a shelter coordinator, I want to reserve a batch for a specific pickup window, so that no other shelter claims the same food.                         | Highest  | 8      |
| FBSR-12 | 2.4 Generate verification code  | FBSR-2 | As a shelter coordinator, I want a unique pickup code when I reserve, so that the donor can confirm my identity at collection.                           | High     | 5      |
| FBSR-13 | 3.1 Validate pickup code        | FBSR-3 | As a donor partner, I want to validate the shelter's code at handover, so that I can confirm the right shelter collected the batch.                      | High     | 5      |
| FBSR-14 | 3.2 Immutable audit log         | FBSR-3 | As a platform administrator, I want every listing, reservation, expiry and collection recorded permanently, so that we can trace who handled each batch. | Medium   | 8      |
| FBSR-15 | 3.3 Donor impact summary        | FBSR-3 | As a donor partner, I want a monthly summary of food I redistributed, so that I can report on our sustainability commitments.                            | Low      | 3      |
| FBSR-16 | 4.1 Shelter verification        | FBSR-4 | As a platform administrator, I want to review and approve shelter registrations, so that only legitimate shelters can claim donated food.                | High     | 8      |
| FBSR-17 | 4.2 Proximity push notification | FBSR-4 | As a shelter coordinator, I want a push notification when a large perishable batch is listed within 5 km, so that I can claim it before it expires.      | High     | 5      |

Total backlog estimate: 62 story points.

## 4. Story Point Rationale

Eight-point stories were assigned to items carrying real uncertainty rather than simply more work: FBSR-11 involves concurrent reservation handling to prevent two shelters claiming the same batch, FBSR-14 requires append-only storage guarantees, and FBSR-16 bundles a registration form with an administrator approval workflow. Two-point stories cover straightforward filtering and tagging with well-understood behaviour.

## 5. Sprint 1

Sprint goal: A donor can list a surplus batch and a shelter can find and reserve it with a valid pickup code.

Duration: 1 week. The five stories below deliver the core post, search and reserve path end to end.

| Key     | Story                          | Points |
|---------|--------------------------------|--------|
| FBSR-11 | 2.3 Reserve pickup window      | 8      |
| FBSR-5  | 1.1 Post surplus batch         | 5      |
| FBSR-12 | 2.4 Generate verification code | 5      |
| FBSR-6  | 1.2 Expiry countdown timer     | 3      |
| FBSR-9  | 2.1 Search by distance         | 3      |
|         | Total committed                | 24     |

All five stories were moved through To Do, In Progress and Done, and the sprint was completed with 24 of 24 points delivered.

## 6. Sprint 2

Sprint goal: Complete the handover, audit trail and shelter verification flows.

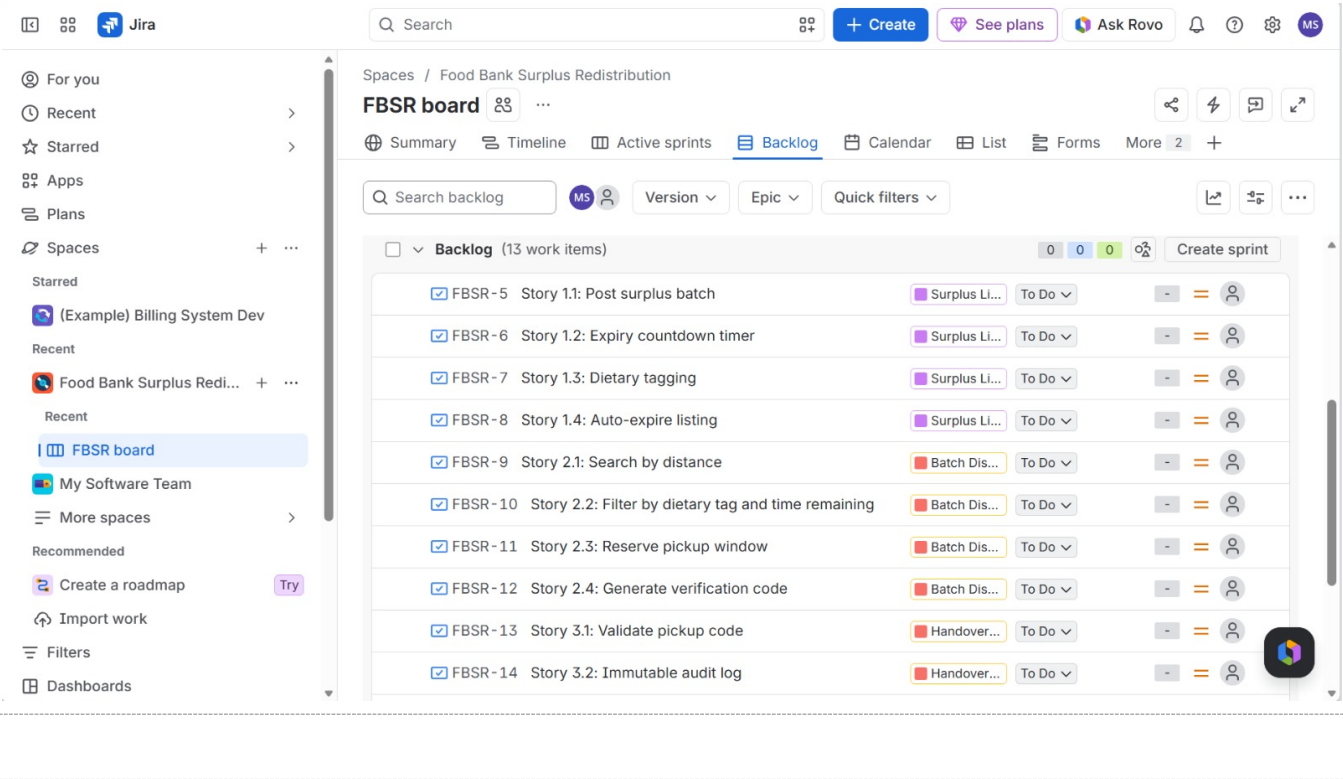
Duration: 1 week. The remaining eight backlog items were committed to this sprint.

| Key     | Story                              | Points |
|---------|------------------------------------|--------|
| FBSR-16 | 4.1 Shelter verification           | 8      |
| FBSR-14 | 3.2 Immutable audit log            | 8      |
| FBSR-13 | 3.1 Validate pickup code           | 5      |
| FBSR-17 | 4.2 Proximity push notification    | 5      |
| FBSR-8  | 1.4 Auto-expire listing            | 5      |
| FBSR-15 | 3.3 Donor impact summary           | 3      |
| FBSR-7  | 1.3 Dietary tagging                | 2      |
| FBSR-10 | 2.2 Filter by dietary tag and time | 2      |
|         | Total committed                    | 38     |

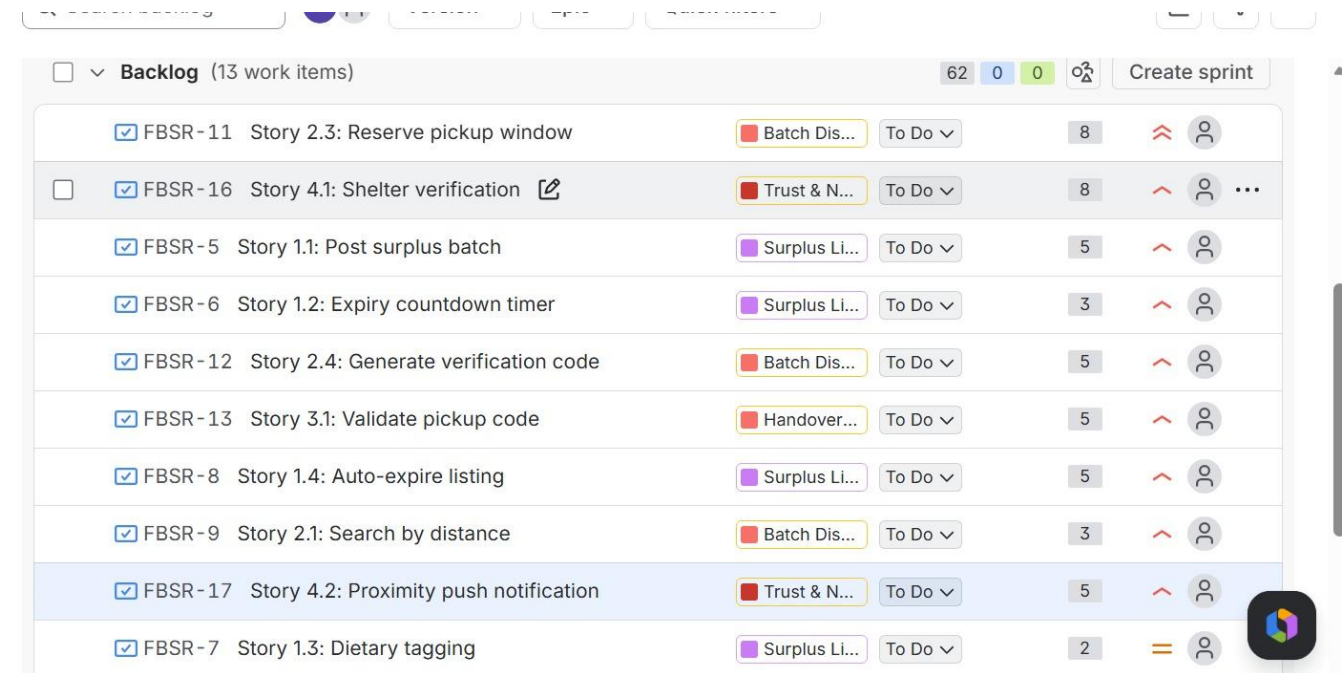
Sprint 2 was completed with 38 of 38 points delivered.

## 7. Screenshots

**Screenshot 1: Jira backlog showing the four Epics and thirteen User Stories, with the Epic panel enabled**



**Screenshot 2: Backlog with priority levels and story point assignments**



|                                     |                                     |                                                       |               |         |   |   |     |
|-------------------------------------|-------------------------------------|-------------------------------------------------------|---------------|---------|---|---|-----|
| <input checked="" type="checkbox"/> | FBSR-12                             | Story 2.4: Generate verification code                 | Batch Dis...  | To Do ▾ | 5 | ^ |     |
| <input checked="" type="checkbox"/> | FBSR-13                             | Story 3.1: Validate pickup code                       | Handover...   | To Do ▾ | 5 | ^ |     |
| <input checked="" type="checkbox"/> | FBSR-8                              | Story 1.4: Auto-expire listing                        | Surplus Li... | To Do ▾ | 5 | ^ |     |
| <input checked="" type="checkbox"/> | FBSR-9                              | Story 2.1: Search by distance                         | Batch Dis...  | To Do ▾ | 3 | ^ |     |
| <input type="checkbox"/>            | <input checked="" type="checkbox"/> | FBSR-17 <u>Story 4.2: Proximity push notification</u> | Trust & N...  | To Do ▾ | 5 | ^ | ... |
| <input checked="" type="checkbox"/> | FBSR-7                              | Story 1.3: Dietary tagging                            | Surplus Li... | To Do ▾ | 2 | = |     |
| <input checked="" type="checkbox"/> | FBSR-10                             | Story 2.2: Filter by dietary tag and time remaini...  | Batch Dis...  | To Do ▾ | 2 | = |     |
| <input checked="" type="checkbox"/> | FBSR-14                             | Story 3.2: Immutable audit log                        | Handover...   | To Do ▾ | 8 | = |     |
| <input checked="" type="checkbox"/> | FBSR-15                             | Story 3.3: Donor impact summary                       | Handover...   | To Do ▾ | 3 | v |     |

**Screenshot 3: Sprint board (Active Sprint view) with work items across To Do, In Progress and Done**

Spaces / Food Bank Surplus Redistribution

## FBSR board ...

Summary Timeline **Active sprints** Backlog Calendar List Forms More 2 +

Complete sprint Group: Story ...

Q Search board Version ▾ Epic ▾ Quick filters ▾

TO DO 2

Story 1.2: Expiry countdown timer

Surplus Listing Management

☒ FBSR-6 3 ^

Story 2.1: Search by distance

Batch Discovery & Reservation

☒ FBSR-9 3 ^

+ Create

IN PROGRESS 2

Story 1.1: Post surplus batch

Surplus Listing Management

☒ FBSR-5 5 ^

Story 2.4: Generate verification code

Batch Discovery & Reservation

☒ FBSR-12 5 ^

DONE 1

Story 2.3: Reserve pickup window

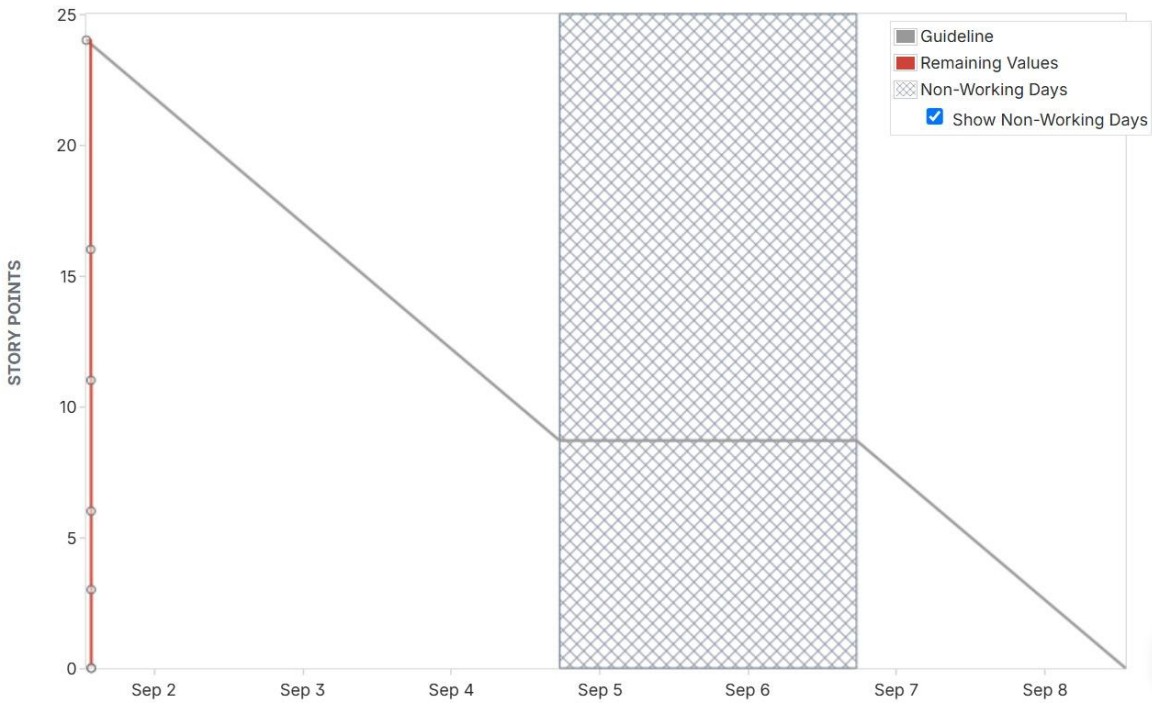
Batch Discovery & Reservation

☒ FBSR-11 8 ^

**Screenshot 4: Burndown Chart — Sprint 1**

# Burndown Chart

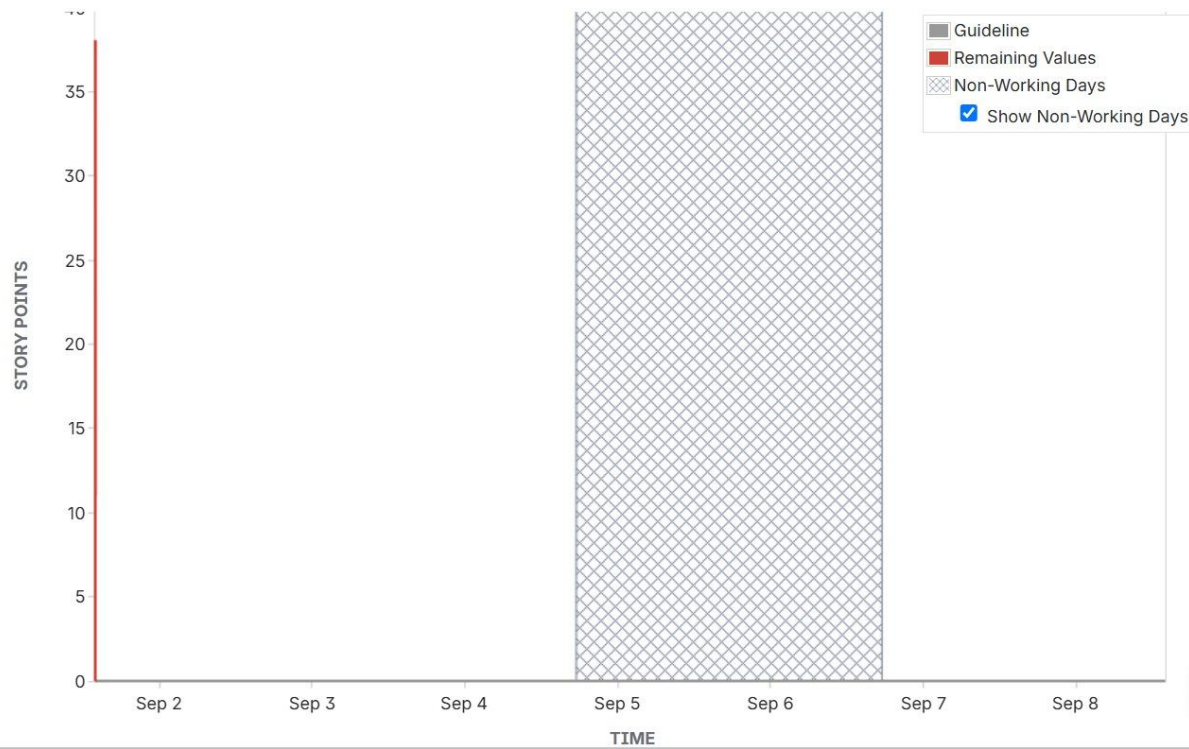
[How to read this chart](#)  



Screenshot 5: Burndown Chart — Sprint 2

# Burndown Chart

How to read this chart





## **8. Reflection**

### **1. Did your estimations reflect the actual effort?**

Because the sprint was simulated within a single session, there is no elapsed-effort data to validate the estimates against, so answer is that the estimates cannot be confirmed by this exercise. The relative spread does appear defensible: the three eight-point stories are the ones with genuine unknowns, while the two-point stories are simple filtering behaviour. If the sprint had run over a real week, I would expect FBSR-16 Shelter verification to have been the weakest estimate, because it combines a registration form and an approval workflow that should have been split into two smaller stories.

### **2. Was your backlog well-prioritized?**

Yes. The backlog was ordered so that the core flow of posting, searching and reserving a batch was delivered first none of the remaining features have value until a shelter can actually claim food. FBSR-11 Reserve pickup window was ranked Highest. FBSR-15 Donor impact summary is reporting rather than core function, so it was ranked Low and scheduled into the second sprint.

### **3. How did your simulated sprint align with your plan?**

All committed points were completed in both sprints, the sprints aligned perfectly with the plan. The burndown charts show why that result is not meaningful: the remaining-value line drops from 24 and from 38 to zero within the first day. The gap between the two lines reflects the compressed simulation rather than the pace of real work.

### **4. What insights did the burndown chart give about your team's capacity?**

The charts do not provide a usable velocity figure, because completing 62 points in one session is not evidence of capacity. What they do tell is the failure mode described in the lab handout: a sharp vertical drop in the remaining-value line indicates work that has not been broken down or paced against real effort. would expect a stepped line tracking close to the guideline, and I would use the first two sprints to establish a velocity baseline before committing to a third.